

BENCHMARK: NON-DETECTION

Results

|                                                   |                               |
|---------------------------------------------------|-------------------------------|
| Mid-Transit Time (Tmid)                           | 2461112.854 +/- 0.022 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.166 +/- 0.047               |
| Transit depth (Rp/Rs)^2                           | 2.75 +/- 1.55 [%]             |
| Orbital Inclination (inc)                         | 87.91 +/- 2.4                 |
| Airmass coefficient 1 (a1)                        | 0.043 +/- 0.023               |
| Airmass coefficient 2 (a2)                        | 2.88 +/- 1.49                 |
| Scatter in the residuals of the lightcurve fit is | 44.58 %                       |
| Best Comparison Star                              | #1 - [415, 370]               |
| Optimal Aperture                                  | 1.58                          |
| Optimal Annulus                                   | 5.0                           |
| Transit Duration (day)                            | 0.051 +/- 0.023               |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                            |
|---------------------------|--------------------------------------------|
| Rp/R■ fit vs published    | 0.166 ± 0.047 vs 0.13272 (deviation 0.59σ) |
| Duration fit vs published | 0.051 d vs 0.0813 d (deviation 1.1σ)       |
| Mid-time O–C              | -7.5 min                                   |
| Depth SNR                 | 3.0                                        |
| Residual scatter          | 44.58 %                                    |

Light curve

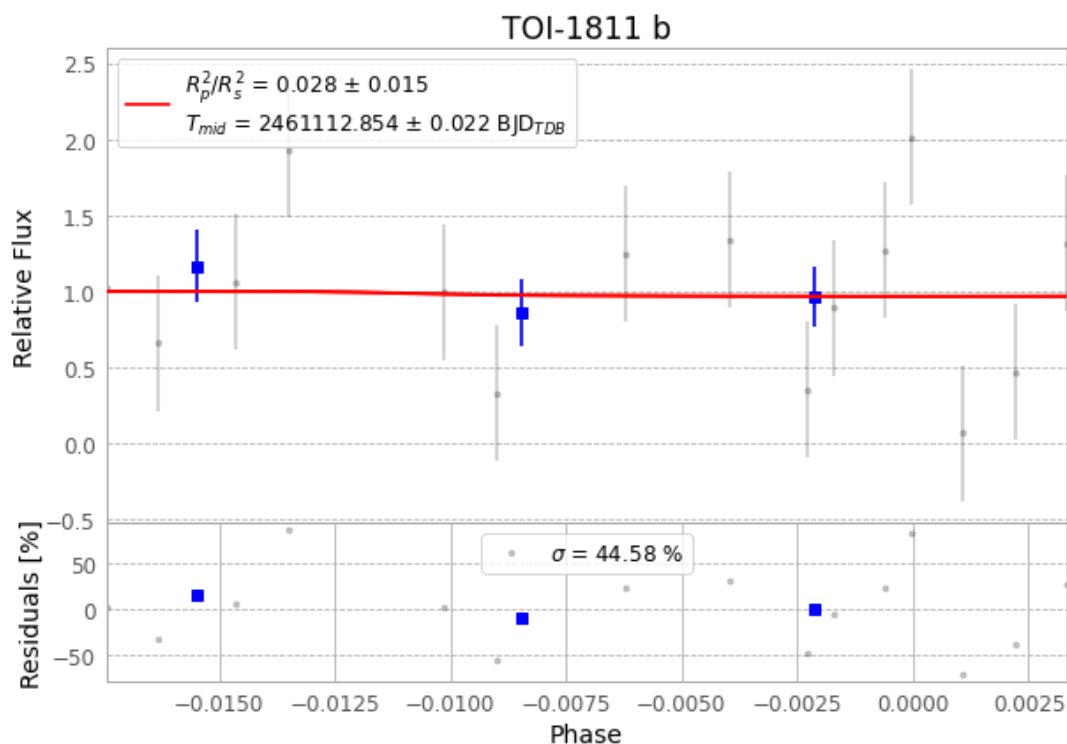


Figure 1 — FinalLightCurve\_TOI-1811 b\_2026-03-12.png (SHA-256 11eb32f0400c6f0a...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [462, 390], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 1f61c4cfeb615df1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit fd74a91

## Data provenance

39 FITS frames (26 MB), TOI1811260313064515.FITS → TOI1811260313083915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1811-260313.log (94f8e3162451...), FinalLightCurve\_TOI-1811 b\_2026-03-12.png (11eb32f0400c...), FinalLightCurve\_TOI-1811 b\_2026-03-12.csv (77d58601a265...)

## Compute resources

Wall time 116.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 23:42Z.